Annual FDI Equity Inflow Forecasting in India: A Time-Ordered Machine-Learning Pilot Study

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Abstract. Foreign direct investment (FDI) equity inflows can support capital formation, productive capacity and cross-border economic partnerships. This paper reports a reproducible annual-frequency pilot for India using the Department for Promotion of Industry and Internal Trade (DPIIT) financial-year series. The data comprise 25 complete annual observations from FY 2000–01 to FY 2024–25, measured in USD million. A one-year lag and a time index are used to compare Linear Regression, degree-two Polynomial Regression with Ridge regularisation, Ridge Regression and Lasso Regression against a persistence forecast. Evaluation uses a chronological hold-out period of six observations. Persistence gives the lowest hold-out RMSE (7,322.4 USD million) and MAPE (11.90%), but its R^2 is negative (-0.580); the small test set does not support a strong generalisation claim. The assigned task is quarterly forecasting with exchange-rate and policy-event predictors. The annual pilot and the supplied cumulative sector spreadsheet do not meet that frequency requirement, so this paper does not claim to complete the quarterly experiment or fabricate quarterly values.

Keywords: foreign direct investment; FDI equity inflow; India; regression; time-series forecasting; Ridge; Lasso; persistence baseline; SDG 8; SDG 17.

1 Introduction

Foreign direct investment is an important channel through which capital crosses national borders. Equity inflows can contribute to business expansion, productive capacity, technology adoption and employment. For India, the movement of FDI equity inflows is of interest to policymakers, analysts and firms because the series reflects investor decisions under changing domestic and international conditions.

The assigned machine-learning problem is to forecast quarterly FDI equity inflows into India. Its intended predictors include historical inflows, USD/INR exchange-rate movements and documented policy-event indicators. Quarterly observations are valuable because they preserve shorter-term changes that annual aggregates can conceal. However, forecasting work is only credible when the target series and predictor observations are aligned at the same frequency and the evaluation respects time order.

The work reported here is an explicitly bounded pilot. The machine-readable DPIIT extract available for modelling in this workflow contains annual financial-year values, not a complete quarterly history. The supplied sector-wise spreadsheet is cumulative from April 2000 to June 2023 and does not carry quarter-level dates. The analysis therefore demonstrates the preprocessing, regression comparison and chronological evaluation workflow at annual frequency while documenting the data needed for the assigned quarterly study.

The objectives are to: (i) describe the long-run annual movement in FDI equity inflows; (ii) compare four regression specifications with a last-observation baseline; (iii) evaluate the models with RMSE, MAPE and R^2 on a time-ordered hold-out; and (iv) specify the exact data gaps that prevent quarterly modelling. The contribution is a reproducible baseline and a transparent audit of the frequency mismatch, not a final quarterly forecast.

2 Related Work and Forecasting Context

The DPIIT fact sheet distinguishes FDI equity inflows from broader FDI aggregates and reports financial-year values in both INR crore and USD million [1]. This distinction matters because total FDI can also include reinvested earnings and other capital components; the target in this pilot is the equity component only. The Open Government Data (OGD) catalogue separately lists an equity-inflow resource described as quarterly [2], while the RBI Database on Indian Economy provides the institutional portal for macroeconomic and exchange-rate series [4].

For forecasting, simple lag-based regression provides an interpretable first benchmark. Polynomial terms can represent curvature, Ridge shrinks coefficients through an L_2 penalty, and Lasso can shrink selected coefficients to zero. A persistence forecast, which carries the most recent observation forward, is a useful reference point: a more complex model should be judged by whether it improves on this simple rule in data that were not used for fitting. Standard forecasting texts emphasise that validation must reflect the temporal structure of the problem [5]. Randomly mixing past and future observations would not represent the intended forecasting use case; chronological evaluation is therefore used here. Time-series cross-validation has also been studied specifically for autoregressive prediction [6].

The available sources support an annual pilot but not a validated quarterly experiment in this working copy. Consequently, the report keeps source description, modelling frequency and forecast claims separate rather than treating catalogue metadata as proof that the underlying quarterly rows were downloaded and checked.

3 Dataset and Preprocessing

3.1 Source and description

The primary source is the DPIIT *Quarterly Fact Sheet on FDI Inflow*, updated through December 2025. Its section on financial-year-wise FDI equity inflow reports annual values in INR crore and USD million. The complete annual observations used here span FY 2000–01 through FY 2024–25 (25 rows). A partial FY 2025–26 observation through December 2025 is excluded because it covers nine months rather than a complete financial year [1].

The user-supplied workbook contains sector-wise cumulative FDI equity totals for 1 April 2000 to 30 June 2023. It is useful for describing the accumulated sector distribution, but it is not a quarterly time series: the workbook does not provide one dated target observation per quarter. It is retained as supplementary context and is not merged into the annual target data.

Table 1: Data fields used in the annual pilot

Field	Definition / role
Financial year	Annual reporting period, FY 2000–01 to FY 2024–25.
FDI equity inflow	Target y_t, USD million, from DPIIT.
Year index	Integer time counter from 0 to 24.
Lag-1 inflow	Previous financial-year target, y_{t-1}.

3.2 Preprocessing and feature construction

Rows are ordered by financial year. A monotonically increasing time index is created, and the one-period lag is calculated by shifting the target down by one row. The first row has no prior-year value and is removed from the modelling table. This leaves 24 annual input–target pairs, from FY 2001–02 through FY 2024–25. No random shuffling is performed. The annual target values are complete in the extracted

series; no target imputation is applied.

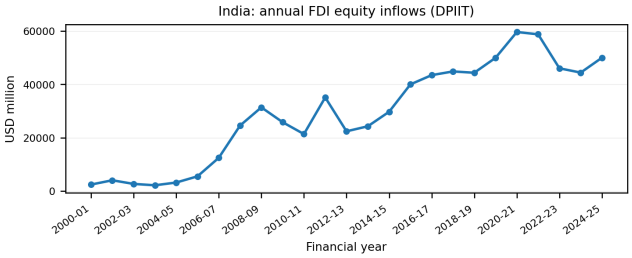


Figure 1: Annual FDI equity inflows in USD million, FY 2000–01 to FY 2024–25. Source: DPIIT [1].

The annual series rises substantially over the full period, but the path is not monotonic. There are visible reversals and periods of acceleration. A long-run trend alone is therefore unlikely to describe every annual change. The graph is descriptive; it does not establish the cause of a rise or fall.

3.3 Frequency and coverage audit

The OGD catalogue contains a resource title describing FDI equity inflows from 2000 through the latest quarter and classifies the resource as quarterly [2]. A Dataful mirror describes a year- and month-wise resource with 504 records through FY 2025–26 [3]. The full underlying quarter/month payload was not retrieved and validated in this analysis. The quarterly labels and record-count metadata are not substituted for the actual rows. Matching quarterly USD/INR exchange rates and dated policy indicators are also not present in the input dataset.

4 Methodology

4.1 Predictors and target

Let y_t denote annual FDI equity inflow in USD million for financial year t. The predictor vector is

$$\mathbf{x}_t = [y_{t-1},\ t],$$

where y_{t-1} is the preceding year's inflow and t is the time index. The fitted models estimate y_t from these two inputs. This deliberately small feature set is used because only the annual target series is available in the pilot. Exchange rates and policy indicators are not silently proxied or fabricated.

4.2 Candidate models

Linear Regression estimates a linear mapping from lagged inflow and time index to the current annual inflow. *Polynomial Regression* expands the two inputs to degree-two terms and applies Ridge regularisation (alpha = 1). *Ridge Regression* applies an L_2 penalty (alpha = 10), while *Lasso Regression* applies an L_1 penalty (alpha = 0.05; maximum iterations = 10,000). Standardisation is fitted within each model pipeline using training observations only. The *persistence baseline* predicts the next value using the previous year's observed inflow:

$$\widehat{y}_t^{\,\mathrm{pers}} = y_{t-1}.$$

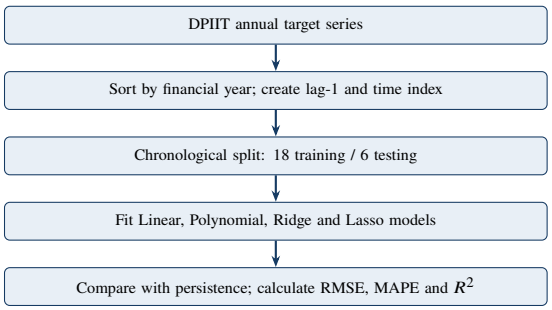


Figure 2: Workflow implemented for the annual-frequency pilot.

4.3 Chronological evaluation design

After the first lagged row is removed, the first 18 observations (FY 2001–02 to FY 2018–19) form the training set. The subsequent six observations (FY 2019–20 to FY 2024–25) form the test set. This is a 75:25 time-ordered split. The test observations occur strictly after the training observations. Model standardisation is estimated on the training data and then applied to the test data through the fitted pipelines.

4.4 Evaluation metrics

Three regression metrics are reported. Root Mean Squared Error measures the typical squared-error scale, with larger errors receiving greater weight:

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - \widehat{y}_i)^2}.$$

Mean Absolute Percentage Error expresses absolute error relative to the observed value:

$$\mathrm{MAPE} = \frac{100}{n}\sum_{i=1}^{n}\left|\frac{y_i - \widehat{y}_i}{y_i}\right|.$$

The coefficient of determination is

$$R^2 = 1 - \frac{\sum_i (y_i - \widehat{y}_i)^2}{\sum_i (y_i - \bar{y})^2}.$$

RMSE is expressed in USD million. Lower RMSE and MAPE are preferred. A negative R^2 indicates that the predictions perform worse on this test set than the constant test-mean benchmark.

5 Experimental Setup and Results

5.1 Implementation

The analysis is implemented in Python using pandas for tabular preparation, scikit-learn pipelines for preprocessing and regression, and Matplotlib for figures. The source script reads the annual CSV, creates the lag and time index, fits each model on the chronological training subset, computes the metrics and regenerates the figures. The model parameters are fixed in the script; this pilot does not claim an extensive hyperparameter search or rolling-origin cross-validation.

Table 2: Model settings used in the annual pilot

Model	Specification
Linear regression	StandardScaler + ordinary least squares.
Polynomial	Degree 2 features + StandardScaler + Ridge ($\alpha = 1$).
Ridge	StandardScaler + Ridge ($\alpha = 10$).
Lasso	StandardScaler + Lasso ($\alpha = 0.05$, max iter 10,000).
Persistence	Previous observed annual inflow (y_{t-1}).

5.2 Hold-out performance

Table 2 reports the chronological test results. The persistence baseline obtains the lowest RMSE (7,322.4 USD million) and the lowest MAPE (11.90%). Ridge has the lowest MAPE among the learned regression models (12.65%), while Linear Regression and Lasso have nearly identical scores. The degree-two Polynomial model has the largest RMSE and MAPE in this comparison.

Table 3: Annual hold-out performance (six observations)

Model	RMSE (USD mn)	MAPE (%)	R^2
Linear regression	7,909.1	14.00	-0.844
Polynomial (degree 2)	8,936.3	15.91	-1.354
Ridge regression	7,999.5	12.65	-0.886
Lasso regression	7,908.9	14.00	-0.844
Persistence (last year)	**7,322.4**	**11.90**	-0.580

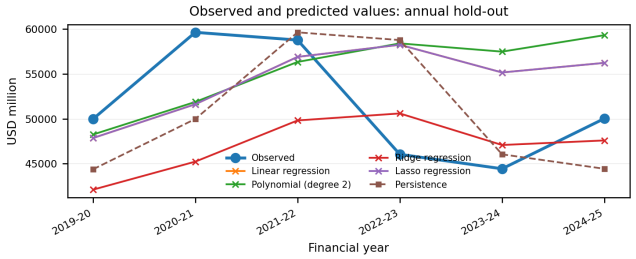


Figure 3: Observed and predicted values on the annual hold-out period.

All five R^2 values are negative. Thus, even though persistence ranks first among the listed approaches, none of the models explains the hold-out variation well under the usual R^2 reference. The six-point test period is small, so the ranking should be treated as descriptive and not as evidence of stable model superiority.

6 Discussion

The annual pilot establishes a reproducible baseline, but its results are modest. The persistence forecast outperforming the fitted models suggests that, for this particular six-year hold-out, the previous annual inflow was a strong simple reference. That does not establish persistence as the best model for quarterly data. The test window includes only six observations, and the annual aggregation hides within-year movements that may be important for short-horizon prediction.

Large changes in FDI can reflect global financing conditions, investor expectations, exchange-rate movements, domestic policy changes and large individual transactions. These influences are not measured directly in the present feature set. The current model therefore describes statistical association in a two-feature annual dataset; it is not a causal model of why FDI rises or falls. A forecast should not be interpreted as an assessment of the success of a particular policy or partnership.

7 Limitations and Requirements for the Quarterly Study

The primary limitation is frequency mismatch. The assigned outcome is quarterly FDI equity inflow, while the modelled DPIIT extract contains annual financial-year totals. The supplied sector workbook is cumulative by sector and period, not a quarter-by-quarter record. Dividing annual values by four would create artificial observations and is not a valid reconstruction of quarterly seasonality or volatility; this was not done.

There are further constraints. First, the annual pilot has only 25 target observations and 24 lagged modelling rows. Second, only six observations are held out for evaluation, making the estimated metrics sensitive to individual years. Third, the predictor set excludes USD/INR exchange rates and policy-event indicators because matching, dated source series were not retrieved and aligned. Fourth, the partial FY 2025–26 total is excluded to avoid comparing nine months with complete financial years. Fifth, annual data cannot support claims about quarter-ahead performance.

To complete the assigned quarterly experiment, the next dataset should contain genuine quarterly FDI equity inflows or a validated monthly series aggregated into consistent quarters. Each quarter should be dated and measured in one consistent unit (USD million or INR crore). A matching RBI USD/INR series should be selected with a clearly stated convention (quarterly average or quarter-end rate). Policy indicators should be encoded from documented event dates, with rules for the quarters in which each event is active. The final experiment should include a larger chronological training, validation and test design, preferably with expanding-window evaluation, and should report all model settings and metrics.

8 SDG Relevance

The project is linked to SDG 17 (Partnerships for the Goals) because FDI represents cross-border investment relationships and can deepen economic partnerships. It is also related to SDG 8 (Decent Work and Economic Growth), since productive investment can support enterprise expansion, capital formation and employment. These links are potential pathways, not automatic outcomes. Inflow volume alone does not establish job creation, decent work or inclusive growth; those claims would require separate employment and sector-level evidence.

9 Conclusion and Future Work

This paper presents a time-ordered annual-frequency pilot of India's FDI equity inflows using 25 complete financial-year observations. Lagged inflow and a time index were used to compare Linear Regression, Polynomial Regression with Ridge regularisation, Ridge, Lasso and a persistence baseline. Persistence achieved the lowest hold-out RMSE and MAPE, but all reported R^2 values were negative and the test set contained only six observations. The results are therefore preliminary and limited to the annual pilot.

The assigned quarterly forecasting task remains incomplete. The next essential step is to retrieve and validate genuine quarterly or monthly FDI equity observations, then align exchange-rate and policy-event predictors before training and evaluating quarterly models. No quarterly accuracy claim is made in this report.

Declarations

Data availability. Annual values are based on the DPIIT fact sheet described in Reference [1]. The supplementary sector workbook was supplied by the student and contains cumulative sector totals. The processed annual CSV is included with the project files.
Code availability. The Python analysis script and figures are included in the accompanying project repository.
Conflict of interest. None declared.
Funding. No external funding declared.
AI assistance. AI assistance was used to organise and format the report and to prepare the reproducible analysis workflow. The student should verify the source data, calculations and interpretation before submission and remains responsible for the final work.

References

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Appendix A. Annual Dataset Used in the Pilot

Table A1 lists the complete annual target series used in this pilot. Values are in USD million and correspond to full financial years only. The modelling table adds the one-year lag and removes the first observation because its lag is undefined.

Table 4: DPIIT annual FDI equity inflows used in the pilot

Financial year	FDI equity inflow (USD mn)	Financial year	FDI equity inflow (USD mn)
2000–01	2,463	2013–14	24,299
2001–02	4,065	2014–15	29,737
2002–03	2,705	2015–16	40,001
2003–04	2,188	2016–17	43,478
2004–05	3,219	2017–18	44,857
2005–06	5,540	2018–19	44,366
2006–07	12,492	2019–20	49,977
2007–08	24,575	2020–21	59,636
2008–09	31,396	2021–22	58,773
2009–10	25,834	2022–23	46,034
2010–11	21,383	2023–24	44,423
2011–12	35,121	2024–25	50,018
2012–13	22,423		

Report status: annual-frequency pilot. The original assignment requires quarterly forecasting; this document must not be submitted or described as a completed quarterly model until the required quarterly target and predictor data have been retrieved, validated and analysed.